

# Gradient Descent

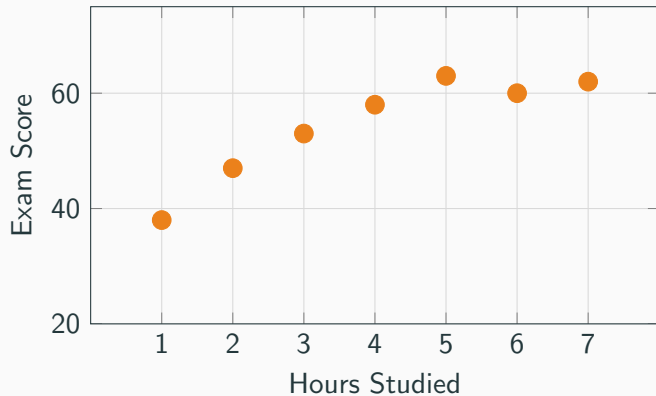
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Hoa T. Vu

# Curve Fitting

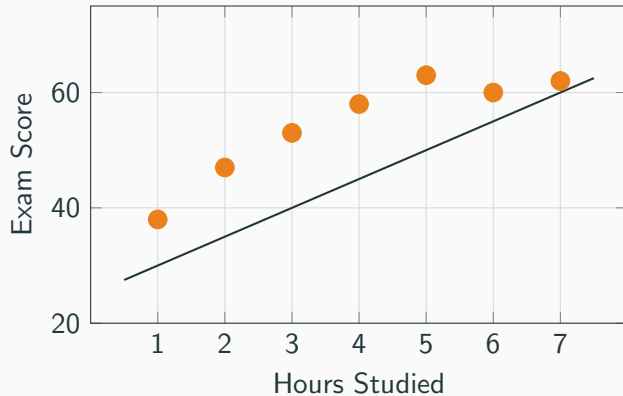
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## Data: Hours Studied vs. Exam Score

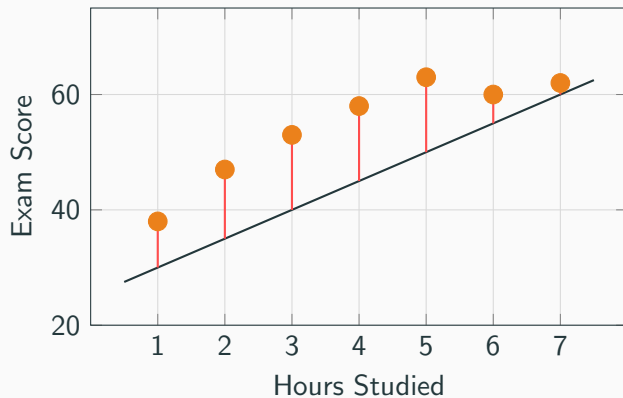


| Hours | Score |
|-------|-------|
| 1     | 38    |
| 2     | 47    |
| 3     | 53    |
| 4     | 58    |
| 5     | 63    |
| 6     | 60    |
| 7     | 62    |

## What Does the Error Look Like?



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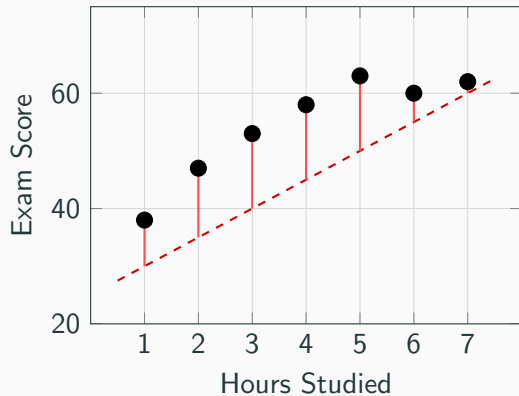
**Residual:**

$$e_i = y_i - \hat{y}_i$$

**Loss:**

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n e_i^2$$

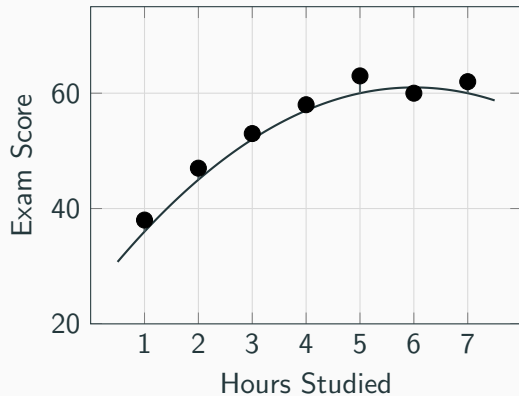
Curve 1 (Linear):  $\hat{y} = 25 + 5x$



| $x$        | $y$ | $\hat{y}$ | $e$ | $e^2$ |
|------------|-----|-----------|-----|-------|
| 1          | 38  | 30        | 8   | 64    |
| 2          | 47  | 35        | 12  | 144   |
| 3          | 53  | 40        | 13  | 169   |
| 4          | 58  | 45        | 13  | 169   |
| 5          | 63  | 50        | 13  | 169   |
| 6          | 60  | 55        | 5   | 25    |
| 7          | 62  | 60        | 2   | 4     |
| $\sum e^2$ |     |           |     | 744   |

$$\mathcal{L} = \frac{744}{7} \approx 106.3$$

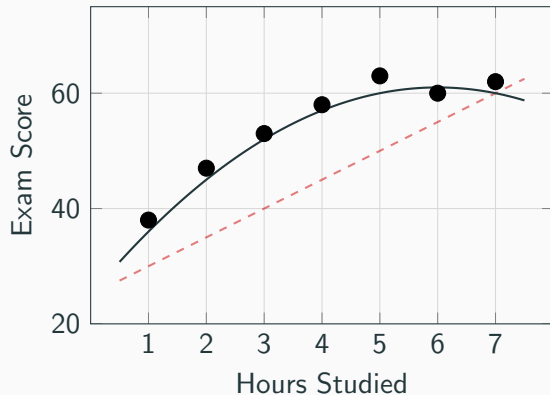
## Curve 2 (Quadratic): $\hat{y} = 25 + 12x - x^2$



| $x$        | $y$ | $\hat{y}$ | $e$ | $e^2$ |
|------------|-----|-----------|-----|-------|
| 1          | 38  | 36        | 2   | 4     |
| 2          | 47  | 45        | 2   | 4     |
| 3          | 53  | 52        | 1   | 1     |
| 4          | 58  | 57        | 1   | 1     |
| 5          | 63  | 60        | 3   | 9     |
| 6          | 60  | 61        | -1  | 1     |
| 7          | 62  | 60        | 2   | 4     |
| $\sum e^2$ |     |           |     | 24    |

$$\mathcal{L} = \frac{24}{7} \approx 3.4$$

# Comparison



**Curve 1** (linear)

$$\hat{y} = 25 + 5x$$

$$\mathcal{L}_1 \approx 106.3$$

**Curve 2** (quadratic)

$$\hat{y} = 25 + 12x - x^2$$

$$\mathcal{L}_2 \approx 3.4$$



# The Optimization Problem

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta)$$

$\theta$  = curve coefficients, e.g.  $(a, b, c)$  for  $\hat{y} = ax^2 + bx + c$ .

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## Key Insight

Fitting a curve to data **is** an optimization problem.

# Partial Derivatives

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# The Idea

$\frac{\partial f}{\partial x}$ : how  $f$  changes as  $x$  varies,  $y$  *fixed*:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

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**In practice:** differentiate w.r.t.  $x$ , treat  $y$  as a constant.  $\frac{\partial f}{\partial y}$  is symmetric.

$\frac{\partial f}{\partial x}$  is the slope of  $f$  along the  $x$ -direction:

$$g(x) = f(x, y_0) \quad \Rightarrow \quad \frac{\partial f}{\partial x} = g'(x)$$

## Examples

**Ex 1.**  $f(x, y) = x^2 + 3xy$

**Ex 3.**  $f(x, y, z) = x^2z + \sin(yz)$

**Ex 2.**  $f(x, y) = e^{xy} + y^2$

## Examples

**Ex 1.**  $f(x, y) = x^2 + 3xy$

$$\frac{\partial f}{\partial x} = 2x + 3y$$

$$\frac{\partial f}{\partial y} = 3x$$

**Ex 2.**  $f(x, y) = e^{xy} + y^2$

$$\frac{\partial f}{\partial x} = y e^{xy}$$

$$\frac{\partial f}{\partial y} = x e^{xy} + 2y$$

**Ex 3.**  $f(x, y, z) = x^2z + \sin(yz)$

$$\frac{\partial f}{\partial x} = 2xz$$

$$\frac{\partial f}{\partial y} = z \cos(yz)$$

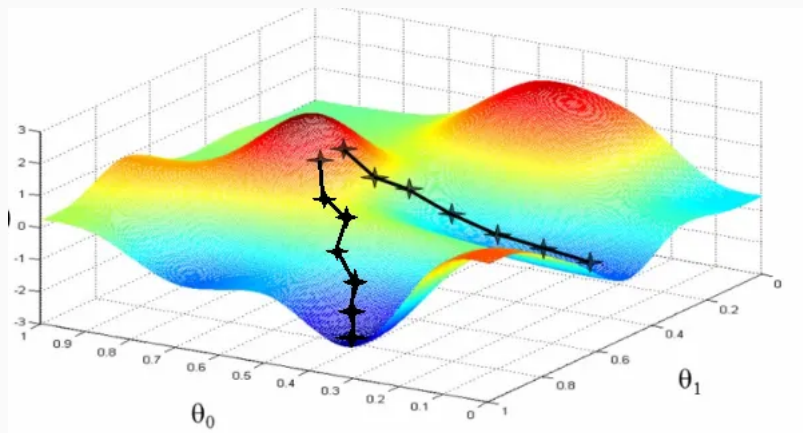
$$\frac{\partial f}{\partial z} = x^2 + y \cos(yz)$$



# Gradient Descent

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# Gradient Descent



# The Gradient

Vector of all partial derivatives of  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ :

$$\nabla f(\theta) = \begin{pmatrix} \frac{\partial f}{\partial \theta_1} \\ \vdots \\ \frac{\partial f}{\partial \theta_n} \end{pmatrix}$$

# The Gradient

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Points toward **steepest ascent** — gradient descent steps the **opposite** way.

## Gradient Examples

**Example 1.**  $f(x) = x^2$

$$\nabla f = \frac{df}{dx} = 2x$$

At  $x = 3$ : gradient = 6

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$$\nabla f = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

At  $(1, 2)$ : gradient =  $(2, 4)^\top$

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At  $(1, 2)$ : gradient =  $(2, 4)^\top$

**Example 3.**  $f(x, y) = 3x^2 - xy$

$$\nabla f = \begin{pmatrix} 6x - y \\ -x \end{pmatrix}$$

At  $(1, 2)$ : gradient =  $(4, -1)^\top$

## Exercise: Compute the Gradient

1.  $f(x, y) = x^3 + 2x^2y - y^3$

2.  $f(x, y) = \ln(x^2 + 1) + e^{xy}$



## Exercise: Compute the Gradient

1.  $f(x, y) = x^3 + 2x^2y - y^3$

$$\nabla f = \begin{pmatrix} 3x^2 + 4xy \\ 2x^2 - 3y^2 \end{pmatrix}$$

2.  $f(x, y) = \ln(x^2 + 1) + e^{xy}$

$$\nabla f = \begin{pmatrix} \frac{2x}{x^2 + 1} + ye^{xy} \\ xe^{xy} \end{pmatrix}$$

Verify by evaluating at a specific point.

# The Update Rule

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta)$$

$\alpha > 0$  learning rate (step size)

$\nabla_{\theta} \mathcal{L}$  gradient, direction of steepest *ascent*

**Example:**  $f(x, y) = e^x + (x - 2)y$

**Partial derivatives:**

$$\frac{\partial f}{\partial x} = e^x + y, \quad \frac{\partial f}{\partial y} = x - 2$$

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**Partial derivatives:**

$$\frac{\partial f}{\partial x} = e^x + y, \quad \frac{\partial f}{\partial y} = x - 2$$

Starting point:  $(x_0, y_0) = (0, 0)$

Initial value:  $f(0, 0) = e^0 + (0 - 2) \cdot 0 = 1$

Learning rate:  $\alpha = 0.1$

## Update 1

**Gradient at  $(0, 0)$ :**

$$\nabla f|_{(0,0)} = \begin{pmatrix} e^0 + 0 \\ 0 - 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

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**Step:**

$$x_1 = 0 - 0.1 \cdot 1 = -0.1, \quad y_1 = 0 - 0.1 \cdot (-2) = 0.2$$

## Update 1

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**Step:**

$$x_1 = 0 - 0.1 \cdot 1 = -0.1, \quad y_1 = 0 - 0.1 \cdot (-2) = 0.2$$

**New value:**

$$f(-0.1, 0.2) = e^{-0.1} + (-0.1 - 2)(0.2) \approx 0.905 - 0.420 = \mathbf{0.485}$$

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$$f_0 = 1 \longrightarrow f_1 \approx 0.485$$



## Update 2

**Gradient at  $(-0.1, 0.2)$ :**

$$\nabla f|_{(-0.1, 0.2)} = \begin{pmatrix} e^{-0.1} + 0.2 \\ -0.1 - 2 \end{pmatrix} \approx \begin{pmatrix} 1.105 \\ -2.1 \end{pmatrix}$$

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**Step:**

$$x_2 = -0.1 - 0.1 \cdot 1.105 = -0.210, \quad y_2 = 0.2 - 0.1 \cdot (-2.1) = 0.410$$

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**Gradient at  $(-0.1, 0.2)$ :**

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**New value:**

$$f(-0.210, 0.410) = e^{-0.210} + (-2.210)(0.410) \approx 0.810 - 0.906 = -\mathbf{0.096}$$

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**Gradient at  $(-0.1, 0.2)$ :**

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**New value:**

$$f(-0.210, 0.410) = e^{-0.210} + (-2.210)(0.410) \approx 0.810 - 0.906 = -\mathbf{0.096}$$

$$1 \longrightarrow 0.485 \longrightarrow -0.096 \quad (\text{function is decreasing})$$

# Pseudo-code

```
initialize theta
set learning rate alpha

repeat until convergence:
    g = gradient of L w.r.t. theta
    theta = theta - alpha * g
```

## Gradient for Quadratic Fitting

Fitting  $\hat{y} = ax^2 + bx + c$ ,  $\theta = (a, b, c)$

**Gradients of MSE loss:**

$$\frac{\partial \mathcal{L}}{\partial a} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i^2$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$$

$$\frac{\partial \mathcal{L}}{\partial c} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

**Update rule:**  $\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}$

# Learning Rate Scheduling

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# Why Schedule the Learning Rate?

**Too large** fast progress, but overshoots near the minimum

**Too small** stable, but very slow

**Schedule** start large, decay over time



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$$\alpha_t = \text{schedule}(\alpha_0, t)$$

# Step Decay

Multiply  $\alpha$  by  $\gamma < 1$  every  $k$  epochs:

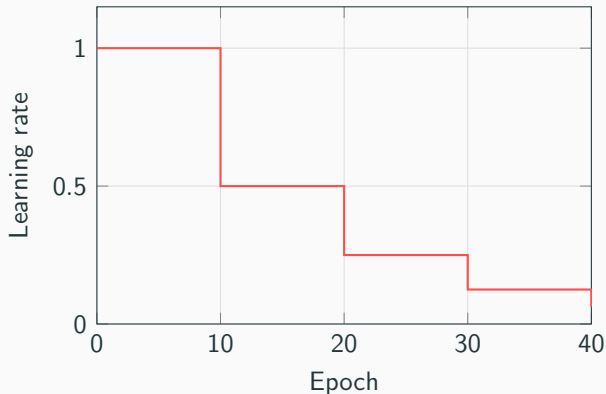
$$\alpha_t = \alpha_0 \cdot \gamma^{\lfloor t/k \rfloor}$$

$\alpha_0$  initial rate

$\gamma$  decay factor (e.g. 0.5)

$k$  step size (e.g. 10)

Large drops at fixed intervals.  
Simple, widely used.



# Exponential Decay

Multiply  $\alpha$  by  $\gamma$  after every epoch:

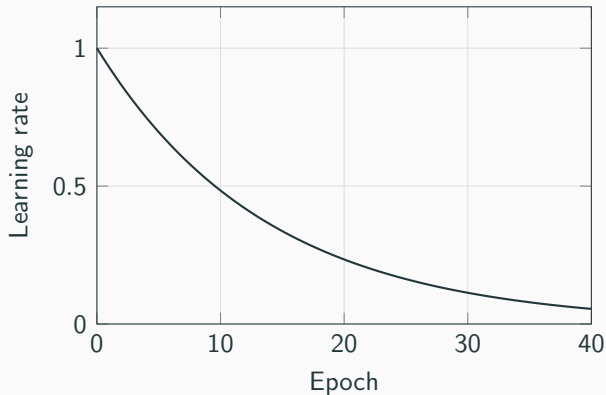
$$\alpha_t = \alpha_0 \cdot \gamma^t$$

$\alpha_0$  initial rate

$\gamma$  decay factor (e.g. 0.93)

Smooth, continuous decay.

Rate never reaches zero.



# Cosine Annealing

Follow a half-cosine curve over  $T$  epochs:

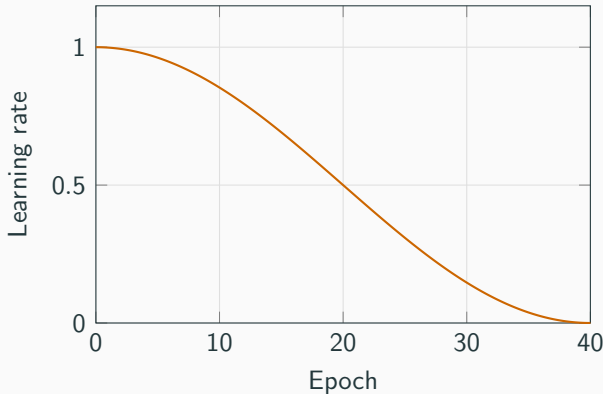
$$\alpha_t = \frac{\alpha_0}{2} \left( 1 + \cos \frac{\pi t}{T} \right)$$

$\alpha_0$  initial rate

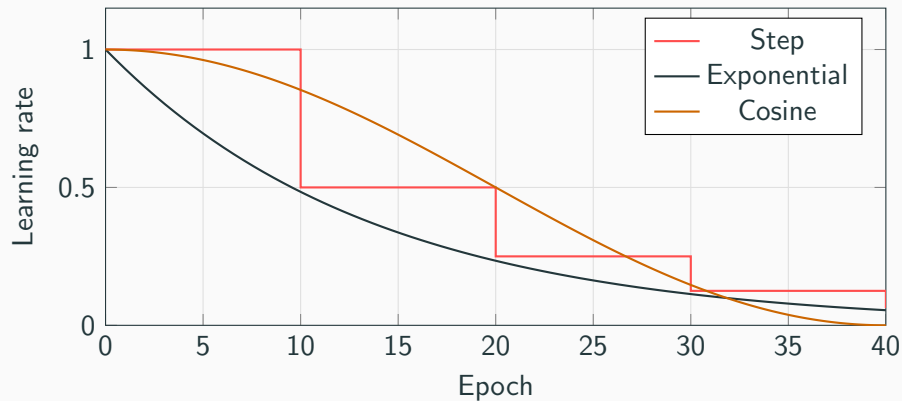
$T$  total epochs

Slow start, fast middle, soft landing.

Very popular in deep learning.



## Comparison



## Convergence Conditions (Robbins–Monro)

$\{\alpha_t\}$  must satisfy:

1.  $\sum_{t=0}^{\infty} \alpha_t = \infty$

reach any point

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**Example** ( $c = 1$ ,  $p = 1$ , harmonic):  $\alpha_t = \frac{1}{t}$ ,  $\sum \frac{1}{t} = \infty$  (cond. 1)  $\sum \frac{1}{t^2} = \frac{\pi^2}{6} < \infty$   
(cond. 2)



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## Fixed learning rate

Satisfies condition 1, but violates condition 2. Converges only to a neighborhood of the minimum.